

HPRCv2-IBD: Identity-by-Descent Detection and Local Ancestry Inference from Pangenome Alignments

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Abstract

Pairwise sequence identity from haplotype-resolved assemblies directly estimates $1 - 2\mu t$, the theoretically optimal quantity for genetic relatedness. We present HPRCv2-IBD, a Rust toolkit that performs IBD detection, local ancestry inference, and relatedness estimation directly from pangenome alignments by applying Hidden Markov Models to sequence identity computed via `img`. Validation on 223 Human Pangenome Reference Consortium individuals across chromosomes 10, 11, 12, and 20 uses matched reference panels and gold-standard VCF-based tools (hap-ibd, RFMix). For IBD, 11/11 (100%) hap-ibd pairs rank in the top 10% by robust statistics across six genomic regions, with 5/6 regions achieving perfect precision-recall at calibrated thresholds. The identity gap between IBD and non-IBD pairs ($\Delta\mu \approx 0.003$) is reproducible across all chromosomes. Eight independent algorithmic improvements converge to the same performance ceiling, establishing a measurement rather than modeling limitation. For local ancestry, a 3-way (EUR/AFR/AMR) pairwise contrast model achieves 76.45% per-haplotype concordance with RFMix across 100 comparisons; the dominant ancestry reaches $F1 = 0.815$ while minority ancestries reflect the fundamental per-window signal-to-noise limitation. Simulation with known ground truth achieves 97.95% concordance, exceeding RFMix (95.9%) by 2.05 percentage points; simulated IBD reaches pair-level $F1 = 0.514$, competitive with hap-ibd (0.489). Grandparent painting in the Platinum pedigree (CEPH 1463) achieves 99.53% mean accuracy from assembly alignments alone. These results establish pangenome identity as the most direct route to the TMRCA signal that coalescent theory identifies as optimal.

1 Introduction

Genetic relatedness between individuals is fundamentally determined by the time to their most recent common ancestor (TMRCA). Under the coalescent Kingman [1982], sequence identity between two haplotypes is a direct estimator of this quantity: $I = 1 - 2\mu t$, where μ is the per-base mutation rate and t is the TMRCA. This relationship is exact to first order under the infinite-sites model and makes pairwise identity a sufficient statistic for TMRCA at each genomic locus. Four independent research programs have recently converged on the conclusion that TMRCA-based relatedness is superior to genotype-based measures: the expected GRM (eGRM) computed from ancestral recombination graphs outperforms the canonical GRM in 97.5% of simulated association settings Fan et al. [2022]; the Branch GRM derived from genealogical trees provides a theoretically optimal measure of genetic relatedness Lehmann et al. [2026]; GeSi classifies relatedness estimators into “full” measures (eGRM, Branch GRM) that capture complete genealogical information versus “shallow” measures (standard GRM, KING) that lose deep coalescent signal GeSi Consortium [2025]; and Super Admixture connects admixture proportions directly to the coancestry matrix $\Theta = Q'\Lambda Q$, where per-population identity means encode the coancestry parameter Chen and Storey [2025].

Despite this theoretical consensus, accessing the TMRCA signal has required either computationally expensive ancestral recombination graph (ARG) inference or discrete genotype data processed through variant-calling pipelines. Current state-of-the-art methods for identity-by-descent (IBD) detection—hap-ibd Browning and Browning [2020], IBDseq Browning and Browning [2013], GERMLINE Gusev et al. [2009]—and local ancestry inference (LAI)—RFMix Maples et al. [2013], FLARE Browning and Browning [2023]—rely on phased genotype data from SNP arrays or short-read sequencing. These approaches introduce three systematic distortions of the underlying TMRCA signal: (i) ascertainment bias from pre-selected marker sets that capture $\sim 85\%$ of common variation in European populations but only 55–65% in African populations Lachance and Tishkoff [2013]; (ii) blindness to structural variants, which contribute ~ 20 Mb of pairwise differences compared to ~ 3 Mb from SNPs Nurk et al. [2022]; and (iii) switch errors from statistical phasing that fragment true IBD segments and degrade with distance from heterozygous sites.

The Human Pangenome Reference Consortium (HPRC) has produced high-quality haplotype-resolved assemblies for hundreds of diverse human genomes Liao et al. [2023]. These assemblies provide chromosome-scale phasing resolved by the assembly process itself, and pangenome graphs constructed via minigraph-cactus Hickey et al. [2024] capture the full spectrum of genomic variation. Tools such as `imp` Garrison et al. [2024] enable efficient computation of pairwise sequence identity directly from pangenome alignments. Crucially, pairwise identity from assemblies is a more direct estimator of $1 - 2\mu t$ than any VCF-derived quantity: it captures all variation simultaneously—SNPs, indels, and structural variants—without ascertainment bias, reference bias, or phasing error. The per-window identity matrix across n haplotypes is, in effect, a local expected GRM estimated directly from assemblies, without requiring VCF generation, statistical phasing, or ARG inference.

In this work, we present HPRCv2-IBD (`impop_k`), a Rust toolkit for IBD detection, local ancestry inference, and relatedness estimation directly from pangenome alignments. A two-state HMM on windowed sequence identity detects IBD segments, an N -state HMM with pairwise contrast emissions infers local ancestry, and Jacquard coefficient estimation enables relatedness analysis—all from the same identity observations. Operating on whole-genome alignments provides several structural advantages: chromosome-scale phasing resolved by the assembly process, eliminating switch errors; capture of all variation simultaneously—SNPs, indels, and structural polymorphisms—without the information loss of variant calling; freedom from marker ascertainment bias, with detection power scaling with coalescent diversity rather than genotyping platform coverage; and a unified framework grounded in coalescent theory.

We validate against hap-ibd and RFMix v2 on chromosomes 10, 11, 12, and 20 using 223 HPRC individuals. Simulation achieves 97.95% ancestry concordance, exceeding RFMix (95.9%) by 2.05 pp; grandparent painting in the Platinum pedigree achieves 99.53% mean accuracy from assembly alignments alone.

2 Related Work

2.1 IBD Detection Methods

Identity-by-descent detection from genotype data is a well-established field. GERMLINE Gusev et al. [2009] pioneered fast seed-and-extend IBD detection using hashed haplotype segments. IBDseq Browning and Browning [2013] introduced a likelihood-ratio framework comparing IBD and non-IBD models per site. hap-ibd Browning and Browning [2020] combined positional Burrows-Wheeler transforms with HMM-based scoring, achieving biobank-scale performance. iLASH Shemirani et al. [2021] uses locality-sensitive hashing for mega-scale IBD detection in diverse populations, while RaPID Naseri et al. [2019] employs random projections on PBWT for ultra-fast IBD calling. Most

recently, PBML authors [2026] scales PBWT-based IBD detection to biobank cohorts via kL-SMEMs (set-maximal exact matches of length $\geq k$ shared by $\geq L$ haplotypes), achieving $4.6\times$ speedup over μ -PBWT. In the pathogen genomics domain, hmmibd-rs authors [2026] reimplements hmmIBD in Rust with Rayon parallelism for *Plasmodium* IBD detection, achieving $100\times$ speedup over the C original. All these methods operate on phased VCF genotype data, requiring variant calling and statistical phasing as preprocessing steps. No assembly-based or pangenome-based IBD detection tool has been published.

2.2 Local Ancestry Inference

Local ancestry inference (LAI) has seen rapid development, with all published methods requiring VCF input. ChromoPainter Lawson et al. [2012] pioneered HMM-based chromosome painting using the Li & Stephens copying model, and SparsePainter Wang and Lawson [2024] extended this to biobank scale using PBWT-based sparse haplotype matching. RFMix Maples et al. [2013] uses conditional random forests on local haplotype windows and remains the most widely used reference method. FLARE Browning and Browning [2023], from the same group that developed hap-ibd, extends the Li & Stephens model Li and Stephens [2003] with composite reference haplotypes for fast and accurate LAI. Gnomix Hilmarsson et al. [2021] combines a base CRF with a neural network smoother. More recently, Recomb-Mix Zhou and Song [2025] achieves very fast inference via graph collapsing of recombination graphs, reducing complexity from $O(nm)$ to $O(np)$ where p is the number of populations. Orchestra Hou et al. [2025] applies CNN and attention layers for 35-population classification but requires days of training on HPC infrastructure.

The only other assembly-based LAI method is ntRoot [Nikolić et al., 2025], which uses k-mer Bloom filters to assign ancestry at ~ 5 Mb resolution ($R^2 = 0.96$ against reference panels). However, ntRoot operates at $500\times$ coarser resolution than `impop_k` (5 Mb vs 10 kb windows), has no HMM for spatial smoothing, cannot detect IBD, and requires pre-built k-mer databases. Our method, `impop_k`, is the first to perform local ancestry inference with per-window HMM resolution directly from pangenome alignments, representing a methodological shift from discrete genotype comparisons to continuous sequence identity observations.

2.3 Pangenome Alignment Tools

The pangenome infrastructure underlying our approach has matured rapidly. Minigraph-Cactus Hickey et al. [2024] constructs pangenome graphs that capture the full spectrum of structural variation. The implicit pangenome graph tool `imp_g` Garrison et al. [2024] computes pairwise sequence identity via transitive alignment closure using cache-oblivious interval trees. AGC Deorowicz et al. [2023] provides efficient compressed storage of assembled genomes. `rustybam` Vollger [2026] offers Rust-based tools for windowed identity computation from PAF/CIGAR data, serving as an independent validation source.

Recent theoretical work on GBWT haplotype matching Sanaullah et al. [2025] formalizes set-maximal and long-match queries on pangenome graphs, offering a path toward graph-native IBD detection where long shared paths through the graph directly indicate shared haplotypes.

2.4 TMRCA-Based Relatedness

The theoretical convergence on TMRCA-based relatedness (Section 1) motivates per-window identity as a local eGRM. All prior “full” GRM methods Fan et al. [2022], Lehmann et al. [2026], GeSi Consortium [2025] require ARG inference; our approach computes $I = 1 - 2\mu t$ directly from assemblies, bypassing both VCF generation and ARG reconstruction. A transformer for TMRCA

prediction [cxt consortium, 2026] independently validates the identity→TMRCAs mapping. The dimensional reduction from m reference haplotypes to k populations—analogueous to the Li & Stephens model Li and Stephens [2003] at the population level—requires fewer reference haplotypes for stable estimation while sacrificing multi-site haplotype information.

3 Methods: IBS Computation from Pangenome Alignments

Identity-by-state (IBS) values serve as the primary observations for all downstream HMM inference. Unlike conventional IBS estimation from genotype arrays or VCF files, our approach computes pairwise sequence identity directly from pangenome alignments, avoiding the ascertainment bias and information loss inherent in variant calling.

Given a target genomic region and a window size W (default 10 kb), we partition the region into non-overlapping windows. For each window, we query the pangenome alignment to compute pairwise sequence identity between all haplotype pairs in a specified sample subset. Two computation modes are supported: (1) per-window queries via `imp` Garrison et al. [2024], which projects alignments through the pangenome graph structure using the reference coordinate system, and (2) PAF-direct mode (`ibs-from-paf`), which parses CIGAR strings from the full alignment file in a single streaming pass.

Raw output is filtered by: identity cutoff (≥ 0.999), self-pair removal, reference removal, and canonical pair ordering to avoid duplicates. The output is a tab-separated file of per-window, per-pair identity values forming the input to IBD and ancestry HMMs. For relatedness inference, the same IBS observations are extended to estimate Jacquard condensed identity coefficients Jacquard [1974] from the four-haplotype sharing pattern between diploid individuals (Supplementary Section L.1).

4 Methods: IBD Detection via Hidden Markov Model

4.1 Model Formulation

We formulate IBD detection as a two-state Hidden Markov Model. Given windowed sequence identity values from pangenome alignments, the model distinguishes regions of shared recent ancestry (IBD, state 1) from background similarity (non-IBD, state 0).

Emissions. Each state emits identity values $o_t \in [0, 1]$ according to a Gaussian distribution $P(o_t | s_t = s) = \mathcal{N}(o_t; \mu_s, \sigma_s^2)$. The non-IBD mean derives from coalescent theory: $\mu_0 = 1 - \theta$, where $\theta = 4\mu N_e$ is the population-scaled nucleotide diversity. Population-specific π values range from 0.00080 (EAS) to 0.00125 (AFR), yielding expected non-IBD identity of 0.99875–0.99920 (Table 1). The IBD mean is $\mu_1 = 0.9997$ with $\sigma_1 = 0.0005$, reflecting only sequencing and assembly errors.

Table 1: Population-specific nucleotide diversity and expected sequence identity parameters.

Population	π (SNPs/bp)	$E[\text{identity} \text{non-IBD}]$	SNR
AFR	0.00125	0.99875	1.06
EUR	0.00085	0.99915	0.74
EAS	0.00080	0.99920	0.69
CSA	0.00095	0.99905	0.81
AMR	0.00100	0.99900	0.85

The per-window signal-to-noise ratio $\text{SNR}_1 = \Delta\mu / \sqrt{\sigma_0^2 + \sigma_1^2}$ ranges from 0.69 (EAS) to 1.06 (AFR). Reliable detection requires accumulating evidence over $L_{\min} = (3/\text{SNR}_1)^2$ consecutive windows, corresponding to minimum detectable segments of ~ 90 kb (AFR) to ~ 190 kb (EAS). Notably, African populations have the best detection power because the identity gap ($\Delta\mu \approx \theta$) is proportionally larger.

Transitions. The transition matrix is parameterized by expected IBD segment length L (in windows) and entry probability p_{enter} : $P(\text{stay IBD}) = 1 - 1/L$, with population-adaptive scaling reflecting coalescent depth. Distance-dependent transitions ($P(\text{switch} \mid d) = 1 - e^{-\lambda d}$) and recombination-rate-aware transitions via genetic maps improve biological realism (Supplementary Section L.2).

4.2 Parameter Estimation

Emission parameters are estimated hierarchically: (1) k-means clustering into two groups, (2) EM fallback with MAP regularization if clusters are degenerate ($\Delta\mu \leq 0.0005$), and (3) BIC model selection to avoid overfitting a two-component model when the data does not support it. All parameters are then refined via Baum-Welch Baum et al. [1970], which re-estimates emissions and transitions jointly over 20 iterations (convergence details in Supplementary Section L.3). Biological bounds are enforced after each M-step to maintain identifiability ($\mu_0 < \mu_1$) and numerical stability.

4.3 Inference and Segment Extraction

The most likely state sequence is computed via the Viterbi algorithm in log-space; posterior state probabilities $\gamma_t(s)$ are computed via forward-backward. IBD segments are extracted by run-length encoding of the Viterbi path, with boundaries refined using posteriors (extension at $\gamma > 0.5$, trimming at $\gamma < 0.2$). Each segment receives a LOD score $\text{LOD} = \sum_t \log_{10}[P(o_t \mid \text{IBD})/P(o_t \mid \text{non-IBD})]$ following the hap-ibd convention Browning and Browning [2020], and a composite quality score $Q \in [0, 100]$ combining posterior strength, consistency, LOD evidence, and segment length.

An identity floor parameter (default 0.9) filters out alignment-gap windows (identity near zero) before HMM inference, treating them as missing data rather than evidence of non-IBD. This reduces variance from $\sigma \approx 0.29$ (bimodal raw distribution) to $\sigma \approx 0.001$ (aligned regions only). Details of the quality score, identity floor analysis, and batch processing are in Supplementary Section L.2.

5 Methods: Local Ancestry Inference via Multi-State HMM

5.1 Model Formulation

We extend the HMM framework to an N -state model where each hidden state corresponds to one of N reference populations. Given pairwise similarity between a query haplotype and reference panel haplotypes, the model infers the most likely ancestral origin at each genomic window. This performs a dimensional reduction relative to the Li & Stephens copying model Li and Stephens [2003] underlying RFMix Maples et al. [2013]: instead of tracking which individual reference haplotype is copied at each SNP (requiring > 500 haplotypes), we ask which *population* shows highest aggregate identity per window. This collapses the state space from m haplotypes to k populations; empirically, ~ 50 haplotypes suffice versus $> 1,400$ for haplotype-matching methods.

Emission model. For each population k , per-haplotype similarities are aggregated into a single score $f_k(o_t)$ using one of four functions: max (default), mean, median, or top- j averaging. The emission probability uses a softmax parameterized by temperature τ :

$$P(o_t | s_t = k) = \frac{\exp(f_k(o_t)/\tau)}{\sum_{j:f_j > 0} \exp(f_j(o_t)/\tau)} \quad (1)$$

The softmax is the Bayes-optimal posterior under Gaussian discriminant analysis with shared variance, and satisfies Luce’s choice axiom. The temperature τ is estimated adaptively from the median per-window spread in population scores, with theoretical scaling $\tau_{\text{opt}} \propto 1/\sqrt{\ln n}$ by extreme value theory.

Pairwise contrast emissions. As an alternative to joint softmax, each population pair (i, j) is evaluated independently via a Bradley-Terry comparison:

$$P_{ij}(k = i | o_t) = \frac{\exp(f_i/\tau_{ij})}{\exp(f_i/\tau_{ij}) + \exp(f_j/\tau_{ij})} \quad (2)$$

with per-pair adaptive temperature $\hat{\tau}_{ij} = \text{median}_t |f_i(o_t) - f_j(o_t)|$. The per-population emission is obtained by aggregating across all pairwise comparisons: $P(o_t | k) \propto \prod_{j \neq k} P_{kj}(k | o_t)$. This resolves the problem that a single temperature cannot simultaneously be sharp for strong contrasts (AFR–EUR, $F_{ST} \approx 0.1$) and gentle for weak ones (EUR–AMR, $F_{ST} \approx 0.02$). Empirically, pairwise contrast improves simulated ancestry from 95.1% (joint softmax) to 97.95%.

Transitions. The transition matrix is uniform across populations, parameterized by a single switch probability p_{switch} . The initial state distribution is uniform. When a genetic map is provided, p_{switch} is modulated by local recombination rate via the Haldane map function. The switch probability is estimated in a two-pass approach: an initial Viterbi pass with a broad prior, followed by regularization toward $p_{\text{switch}} = 0.001$.

5.2 Automatic Parameter Configuration

Optimal parameters differ substantially between datasets: simulated data with strong population separation requires $w = 0.7$, $ec = 1$, while HPRC real data requires $w = 0.12$, $ec = 15$, with a 9.8 pp cross-application penalty. We introduce automatic configuration based on $D_{\min}$, the minimum Cohen’s d across all reference population pairs:

$$D_{\min} = \min_{i < j} \frac{|\bar{f}_i - \bar{f}_j|}{\sqrt{(\hat{\sigma}_i^2 + \hat{\sigma}_j^2)/2}} \quad (3)$$

The pairwise weight scales with the coefficient of variation of pairwise Cohen’s d values, while the emission context scales inversely with $D_{\min}$: $ec^* = \text{round}(0.1/D_{\min})$, clamped to $[1, 15]$. This formula recovers 3.3 pp on simulated data and eliminates manual parameter tuning.

5.3 Inference

The most likely ancestry sequence is computed via Viterbi decoding; posterior ancestry probabilities via forward-backward. As an alternative, posterior decoding ($\hat{a}_t = \arg \max_k \gamma_t(k)$) makes independent per-window decisions, which is advantageous for detecting short minority ancestry tracts where

Viterbi’s path constraint suppresses isolated switches. The switch probability is refined via Baum-Welch training over all sample sequences simultaneously. Genome-wide admixture proportions, per-population tract lengths, ancestry LOD scores, cross-validation, and output format details are in Supplementary Section L.4.

6 Validation Against Established Methods

We validate IBD detection and local ancestry inference by running pangenome-based and VCF-based methods on the same samples, genomic regions, and reference panels. All concordance metrics are computed from base-pair-level overlap and per-window agreement.

6.1 Design

Validation spans chromosomes 10, 11, 12, and 20, using 223 HPRC individuals present in both the pangenome and phased VCFs. IBD is compared against hap-ibd Browning and Browning [2020] (v1.0, default parameters, 4,091-sample VCFs phased with SHAPEIT5). Local ancestry is compared against RFMix v2 Maples et al. [2013] (default parameters). The matched reference panel design—100% sample overlap between tools—isolates the effect of input data representation (pangenome similarity vs. VCF genotype likelihoods) from panel composition effects. Detailed parameter settings for both tools are in Supplementary Section L.5.

IBD evaluation. The primary metric is ranking-based: all haplotype pairs are ranked by robust statistics (trimmed mean, median) of pangenome identity, and we assess whether known IBD pairs cluster at the top. This avoids threshold sensitivity. The secondary metric is segment-level overlap: Jaccard index, precision, recall, and F1, with hap-ibd as ground truth. Population-stratified metrics are computed separately for intra- and inter-population pairs.

Ancestry evaluation. Per-window concordance $C = W^{-1} \sum_w \mathbb{I}[\hat{a}_w^{\text{ours}} = \hat{a}_w^{\text{RFMix}}]$ is the primary metric. Per-population precision, recall, and F1 assess minority ancestry detection. The chr20 baseline uses a 2-way (EUR/AFR) model with 79 matched reference individuals; the chr10/11/12 extension uses a 3-way (EUR/AFR/AMR) model with 31 HPRC pangenome references (62 haplotypes) for our tool and 1,486 VCF reference haplotypes for RFMix, reflecting the smaller available pangenome panel. Detailed metric definitions, confusion matrices, and switch-point accuracy are in Supplementary Section L.5.

7 Results

7.1 Validation Overview

We validate IBD detection and local ancestry inference against gold-standard VCF-based tools on chromosomes 10, 11, 12, and 20 of the HPRC dataset. Validation uses 223 HPRC individuals with matched reference panels, matched genetic maps, and the same samples for both tools, isolating methodological differences from compositional effects. IBD is compared against hap-ibd Browning and Browning [2020]; local ancestry is compared against RFMix v2 Maples et al. [2013]. Detailed parameter sweeps, per-pair breakdowns, and supplementary analyses are in Supplementary Section K.

7.2 IBD Detection

7.2.1 Baseline Concordance (chr20)

On chr20 (44 sample pairs, 51 hap-ibd segments), the best HMM configuration (identity floor = 0.9, minimum segment length 2 Mb, 10 Baum-Welch iterations) achieves $F1 = \mathbf{0.283}$ (precision 0.317, recall 0.270). Scaling to all 2,850 HPRC pairs confirms the pattern: pair-level recall is 68.8% (11/16 hap-ibd pairs detected) with precision of 4.3% at the default threshold, improving to segment $F1 = 0.432$ at a 3.0 Mb minimum length (Supplementary Section K.6).

The moderate concordance reflects a fundamental signal-to-noise limitation: the identity gap between IBD and non-IBD pairs ($\Delta\mu \approx 0.003$) is dwarfed by alignment-driven variance ($\sigma \approx 0.29$), yielding per-window $SNR \approx 0.01$ —approximately $55\times$ worse than VCF-based methods (Supplementary Section K.3). Eight independent algorithmic improvements (distance-aware transitions, genetic-map transitions, logit-transform emissions, background filtering, z-score normalization, population-adaptive transitions, coverage-ratio auxiliary emission, region exclusion) all converge to the same performance ceiling, establishing that **$F1 \approx 0.28$ represents a measurement limitation, not a modeling limitation** (Supplementary Section K.4).

7.2.2 Multi-Chromosome Ranking-Based Evaluation

To test whether pangenome identity discriminates IBD from non-IBD pairs across diverse genomic contexts, we evaluated all 11 high-confidence hap-ibd pairs ($LOD \geq 3$) across six regions spanning chr10, chr11, and chr12 (Table 2).

Table 2: Multi-chromosome IBD ranking evaluation. For each region, all haplotype pairs are ranked by summary statistic of pangenome identity. “Top 10%” indicates whether the IBD pair ranks in the top decile.

Summary statistic	IBD pairs in top 10%	Ranked #1
Mean	10/11 (91%)	6/11
Trimmed mean (5%)	11/11 (100%)	6/11
Median	11/11 (100%)	6/11

With robust statistics (trimmed mean or median), **all 11 IBD pairs (100%) rank in the top 10%** of pairs by pangenome identity, with **6 ranking #1** in their region. The single outlier under the arithmetic mean (HG01175 vs. HG01192, chr10, rank 103/120) is caused by 5 alignment-artifact windows; robust statistics resolve it to rank 10/120 (top 8.3%). At region-specific identity thresholds, **5 of 6 regions achieve perfect precision and recall** (Supplementary Section K.11).

The identity gap between IBD and non-IBD pairs is remarkably consistent across chromosomes: $\Delta\mu = 0.00315$ (chr10), 0.00289 (chr11), 0.00326 (chr12)—all within 0.003 ± 0.0004 , consistent with the chr20 finding. Segment length strongly discriminates true positives (>3.8 Mb) from false positives (<2.5 Mb).

7.2.3 Distance-Aware Transitions

On multi-chromosome HPRC validation, enabling distance-aware transitions with identity floor 0.998 improves mean pair-level $F1$ from 0.380 to **0.404** (+6.3%), with per-chromosome results of chr11 $F1 = 0.624$ (+7.6%), chr12 $F1 = 0.440$ (+6.8%), and chr10 $F1 = 0.148$ (unchanged). This is now the recommended default for IBD detection.

7.3 Local Ancestry Inference

7.3.1 2-Way Baseline (chr20)

On chr20 with a 2-way EUR/AFR model (HG00733, 30 EUR + 49 AFR reference panel), the `max` emission model achieves **93.2%** concordance with RFMix (EUR F1 = 0.965, AFR F1 = 0.158). Multi-sample validation across 15 AMR individuals yields mean concordance **93.4%** (range 90.2–95.5%), confirming stability. The per-window signal-to-noise ratio for ancestry discrimination is SNR ≈ 0.09 : in 75% of true AFR windows, EUR references have *higher* similarity than AFR references, making correct assignment statistically impossible at single-window resolution. The HMM achieves its concordance through temporal smoothing (transition-derived state persistence), not per-window signal detection (Supplementary Section K.9).

7.3.2 3-Way Concordance (chr10, chr11, chr12)

We extended to a 3-way EUR/AFR/AMR model on chr10, chr11, and chr12, modeling the Native American component present in all AMR query samples. Nine AMR query samples (18 haplotypes) were evaluated across six 1 Mb regions using a matched panel of 31 HPRC individuals (8 EUR, 8 AFR, 15 AMR). A sweep of 13 emission model configurations identified the top-3 averaging model as optimal (Table 3).

Table 3: 3-way ancestry concordance with RFMix (100 haplotype-region comparisons with available data, 6 regions across chr10/11/12). Top-3 emission model.

Ancestry	Precision	Recall	F1
AMR	95.5%	71.1%	0.815
EUR	29.2%	54.9%	0.381
AFR	30.8%	100.0%	0.471
Overall concordance			76.45%
Per-chromosome range			71.9–80.7%
Best region (chr11:25–26 Mb)			88.9%

The dominant ancestry (AMR) is detected with F1 = 0.815; minority ancestries are weaker, reflecting the signal-to-noise limitation of 62 pangenome reference haplotypes versus RFMix’s 1,486 VCF haplotypes. The dominant error pattern is AMR windows misclassified as EUR or AFR (Supplementary Section K.10).

7.4 Simulation Validation

To evaluate under controlled conditions, we constructed a simulation framework using `msprime` to generate known ancestry tracts stitched onto real HPRC assemblies (`pangenome_sim`), preserving realistic pangenome identity characteristics.

7.4.1 Ancestry

With pairwise contrast emissions, `impop_k` achieves **97.95%** concordance with ground truth, *exceeding* RFMix (95.9%) by **2.05 pp** on the same panel. The pairwise contrast model—treating each population pair independently with per-pair adaptive temperature—is the key driver of improvement over the baseline softmax (95.1% $\rightarrow$ 97.95%).

Table 4: Simulated ancestry concordance: `impop_k` vs. RFMix on the same 46-haplotype panel (chr12, 5 individuals, pairwise contrast emissions).

Metric	<code>impop_k</code>	RFMix (46 hap)	RFMix (26 hap)
Concordance	97.95%	95.9%	96.1%
AFR F1	0.982	0.963	0.965
EUR F1	0.977	0.954	0.956
Gap vs. RFMix		+2.05 pp	—

7.4.2 IBD

On a simulated 50-individual pedigree (chr12), `impop_k` achieves pair-level F1 = **0.514**, competitive with hap-ibd’s 0.489 on the same data. Baum-Welch training is essential: without EM iterations, IBD detection fails entirely; with 10 iterations, the HMM converges to functional emission parameters. Panel size is critical: 30 reference haplotypes yield 0% IBD detection, while 50 haplotypes achieve 51.4%.

7.5 Platinum Pedigree Validation (CEPH 1463)

The CEPH 1463 (Platinum) family provides a real-data validation with known family structure: 5 of the 11 offspring have HPRC v2 assemblies, along with all 4 grandparents. We performed 4-state founder painting—assigning each genomic window in each offspring to one of the 4 grandparental origins based on pangenome IBS similarity.

The 4-state model achieves **99.53%** mean concordance across all 5 offspring (range 97.5–99.5%), making this the most convincing validation result. This is a real family with no simulation artifacts, demonstrating that pangenome assembly alignments alone can near-perfectly reconstruct grandparental inheritance patterns. The 8-state model (distinguishing maternal/paternal grandparental haplotypes) drops to 70–85%, reflecting the inherent difficulty of resolving which haplotype within a grandparent was transmitted.

7.6 Computational Performance

We benchmarked two IBS computation modes: the original per-window `impg` pipeline and PAF-direct mode (`ibs-from-paf`).

For a single-chromosome analysis (chr12, 40 reference haplotypes), `ibs-from-paf` completes in **46.3 s** vs. 368 s (7.9× speedup), with peak memory of **101 MB** vs. 2,900 MB (28.7× reduction). Scanning all 22 autosomes, `ibs-from-paf` processes the complete genome in **195 s** (3.3 min), yielding a **1,090×** speedup over the `impg` pipeline (~59 h extrapolated). Memory remains constant at 101 MB regardless of genome coverage.

The complete VCF-based IBD/ancestry pipeline typically requires 5–7 hours per sample. The pangenome pipeline achieves comparable or superior accuracy in 3.3 minutes of marginal compute time per sample pair, given pre-existing alignments—a fundamentally different point in the accuracy–speed trade-off space.

8 Discussion

8.1 Assembly Identity as a TMRCA Estimator

The central finding is that sequence identity from pangenome alignments provides a practical, theoretically grounded estimator of pairwise TMRCA. As established in Section 1, four independent research programs converge on the superiority of TMRCA-based relatedness. Our per-window identity matrix constitutes a local eGRM estimated directly from assemblies, bypassing both VCF generation and ARG inference. Formally, the identity matrix is an affine transformation of the Branch GRM: $G_w = (1 - 2\mu\bar{H}_w)\mathbf{1}\mathbf{1}' + 2\mu \cdot \text{bGRM}_w + E_w$, with spectral equivalence (Davis-Kahan angular deviation $<1^\circ$) ensuring that identity-based PCA reproduces Branch GRM structure. Per-window Fisher information efficiency is 56%; genome-wide aggregation across $\sim 300,000$ windows yields $>99.99\%$ information capture.

This framework explains the three key results. The 97.95% simulated concordance confirms that assembly identity captures the essential TMRCA signal. The Platinum pedigree’s 99.53% mean concordance exploits the strong TMRCA contrast between unrelated grandparents, with only two meioses producing long, easily detected tracts. The remaining gap on real data (76.45%) reflects a measurement limitation—per-window SNR of 0.69–1.06—not a theoretical one.

8.2 Advantages of the Pangenome Approach

Beyond the structural advantages outlined in Section 1 (chromosome-scale phasing, all-variant capture, no ascertainment bias), a notable empirical finding is that IBD detection power scales with coalescent diversity (π) rather than genotyping platform coverage: SNR is highest in African populations (1.06) and lowest in East Asian (0.69), inverting the hierarchy of SNP array-based methods. This makes the approach particularly valuable for under-represented populations Lachance and Tishkoff [2013]. The PAF-direct mode completes a full 22-autosome scan in 195s using 101 MB RAM, making whole-genome analysis practical on a laptop.

8.3 Algorithmic Maturity and the Emission Ceiling

Our benchmarking program (241 benchmarks) reveals that the HMM operates near an empirical ceiling. The transition dominance ratio $R \approx 2,800$ places the system in a regime where emission improvements are bounded: a feature saturation theorem shows any emission transformation changes accuracy by $<0.3\text{pp}$, because only ~ 6 boundary windows per chromosome are sensitive while $\sim 14,000$ are locked by accumulated transition evidence. This explains why 32 independently motivated emission features do not compound. The pipeline retains $\sim 73\%$ of per-window Fisher information about TMRCA. All accuracy claims are grounded in empirical benchmarks; prescriptive predictions about intervention effects achieved 0/8 accuracy due to cascading interactions through auto-configuration (Supplementary Section J.3).

8.4 Limitations

Several limitations should be noted. Per-window identity has a fundamentally low SNR for ancestry discrimination (~ 0.001 difference between populations at 10 kb resolution), making the HMM dependent on temporal smoothing rather than per-window classification. Minority ancestry detection degrades as $\pi_j^{3/2}$ (Supplementary Section J.8), explaining why AMR at 20% has $\sim 1/4$ the detection power of EUR at 50%. The current HPRC panel (46 haplotypes) limits accuracy for under-represented populations, and panel scaling follows $O(1/\ln N)$ —doubly logarithmic in panel size.

For IBD, the per-window identity ceiling limits segment-level concordance with hap-ibd ($F1 = 0.514$), though pair-level detection is strong (68.8% recall, top pairs at Jaccard = 0.953). Identity-based IBD performs worse in high-SV regions due to alignment fragmentation, and logit-space emission transforms, while theoretically improving model adequacy, regress on real data due to bimodal emission artifacts (Supplementary Section J.7).

Assembly quality and pangenome completeness remain prerequisites: regions without aligned segments produce gaps in the observation sequence, and assembly errors can mimic or mask IBD signal.

8.5 Future Directions

Several extensions follow from our findings. Graph-topology features (node sharing, bubble traversal patterns in pangenome graphs) may capture allele-specific information lost in aggregate identity, particularly in SV-rich regions where identity-based detection fails. Larger reference panels from HPRC Release 3 (>350 assemblies) will enable $K \geq 5$ population models and improve accuracy for under-represented groups. The per-window eGRM output enables downstream applications—local PCA, per-window F_{ST} , heritability estimation, and association testing—without VCF generation. Ancestry-specific eGRM Tang and Chiang [2025] conditioned on local ancestry calls provides an assembly-based as-eGRM without ARG inference. IBD segment lengths enable VCF-free demographic inference via established tools (HapNe-IBD, IBDNe), and a multi-pulse EM algorithm extending MultiWaver Ni et al. [2019] is already implemented for ancestry tract-based dating. Finally, including archaic assemblies (Neanderthal, Denisovan) as reference states would enable assembly-based introgression detection.

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Data Availability

HPRC assemblies and pangenome graphs are available from <https://humanpangenome.org>. The HPRCv2-IBD software is available as open-source Rust code.

Code Availability

Source code for all tools (`ibd-cli`, `ibs-cli`, `ibs-from-paf`, `ancestry-cli`, `jacquard-cli`) is available in the HPRCv2-IBD repository. The software includes 7,221 automated tests across 241 validation benchmarks.

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